TEXORA · LECTURE SHEET

Project Planning: Spin Plan

A spin plan is the complete numerical design of a spinning mill for a stated product: starting from the yarn and its daily target, it works backwards through every process — allowing for waste at each stage — to the raw material requirement, and forwards into the number of machines of each type. This sheet builds a full plan for 2000 kg/day of combed Ne 30, using the machine data calculated in Topics 1–5.

YM-3207 · ADVANCED SPINNING CALCULATIONS · TOPIC 8 · SPECIALIZATION — YARN MANUFACTURING

1. The logic of a spin plan

A spin plan answers four questions in order. (1) **What product, how much?** — count, carded or combed, kilograms per day. (2) **What process route and parameters?** — drafts, twists, speeds and production per machine, taken from the kind of calculations in Topics 1–5. (3) **How much material must enter each stage?** — computed backwards from the yarn, dividing by the retention (100 − waste%) of each stage in turn. (4) **How many machines?** — each stage's daily duty divided by one machine's daily production, rounded up. The plan is balanced when every stage's installed capacity slightly exceeds its duty, with the ring frame — the capacity-setting machine — planned first and most exactly.

2. Step 1: the material balance

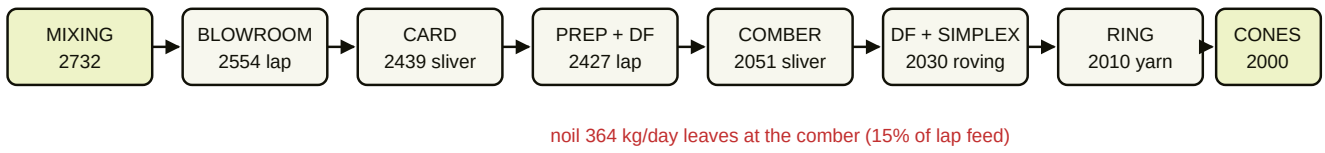
Waste allowances for a combed Ne 30 plan (typical values; a real plan uses the mill's own audited figures — see YM-4101, Topic 11):

STAGE (BACKWARDS)	WASTE	RETENTION	FEED REQUIRED (KG/DAY)
Yarn (target)	—	—	2000.0 delivered
Winding	0.5%	0.995	2000/0.995 = 2010.1
Ring frame	1.0%	0.990	2010.1/0.990 = 2030.4
Simplex	0.5%	0.995	2040.6 (roving)
Finisher drawframe	0.5%	0.995	2050.8 (comber sliver)

Comber	15.5% (15% noil + 0.5%)	0.845	2427.0 (lap)
Breaker drawframe + lap prep	0.5%	0.995	2439.2 (card sliver)
Card	4.5%	0.955	2554.1 (blowroom lap)
Blowroom	6.5%	0.935	**2731.7 (mixing)**

READING THE BALANCE

Mixing required = **2732 kg/day** for 2000 kg/day of yarn.
Overall yarn realisation = 2000 / 2731.7 × 100 = **73.2%**.
The figure looks low next to carded-yarn realisations (high 80s) until the comber is noticed: 15% noil alone caps realisation near 85%, and blowroom plus card waste take most of the rest. The noil is not destroyed value — it is sold or respun into coarse counts — but it is not in this yarn, and the plan must buy cotton for all of it.



Daily mass flow of the plan (kg/day). Each stage's feed is the previous stage's duty.

Figure 1 — The spin plan as a mass-flow chain. Planning runs right to left (from the yarn); production runs left to right.

3. Step 2: machine allocation

Machine productivities are taken from this course's earlier sheets: card 13.0 kg/h delivered sliver (Topic 2, at 85%); comber 41.6 kg/h delivered, taken at 90% → 37.4 kg/h (Topic 3); finisher drawframe 88.1 kg/h per delivery (Topic 4); simplex 0.69 kg/h per spindle (Topic 5); ring 17.6 g/h per spindle (Topic 5); blowroom line 297 kg/h of lap (Topic 1). The mill works 24 h/day.

DEPARTMENT	DAILY DUTY	PER-MACHINE RATE	CALCULATION	INSTALL
Blowroom	2554 kg lap	297 kg/h	2554/297 = 8.6 h/day	**1 line**

Cards	2439 kg sliver	13.0 kg/h	2439/(13.0 × 24) = 7.8	**8 cards**
Combers	2051 kg sliver	37.4 kg/h	2051/(37.4 × 24) = 2.3	**3 combers**
Drawframes (per passage)	≈ 2041–2427 kg	88.1 kg/h/delivery	2427/(88.1 × 24) = 1.15	**2 deliveries per passage**
Simplex	2030 kg roving	0.686 × 96 spindles × 0.90 = 59.3 kg/h per 96-spindle frame	2030/(59.3 × 24) = 1.43	**2 frames**
Ring frames	2000 kg yarn	17.6 g/h per spindle, 92% eff.	2,000,000/(17.6 × 24 × 0.92) = 5160 spindles	**5 frames × 1200 = 6000 spindles**
Autocone	2000 kg cones	2.2 kg cone per 124 min → 25.5 kg/day per drum	2000/25.5 = 78 drums	**80 drums**

WORKED EXAMPLE — THE RING FRAME CALCULATION IN FULL

One spindle makes 17.6 g/h (Topic 5). In 24 h at 92% efficiency it makes 17.6 × 24 × 0.92 = 387.6 g/day.
Spindles required = 2,000,000 g / 387.6 g = **5160 spindles**.
Frames of 1200 spindles: 5160/1200 = 4.3 → **5 frames (6000 spindles)**.
Installed capacity = 6000 × 387.6 g = 2325 kg/day — a 16% margin over target, which absorbs count changes, extra ends down and maintenance stoppages. Margins like this are deliberate: a plan balanced to the last spindle fails the first week it runs.

4. Checking and using the plan

- **Balance check:** every upstream stage's installed daily capacity must meet or exceed the next stage's feed requirement from the mass balance — not merely the yarn target.
- **Bottleneck check:** the stage with the least spare capacity limits the whole mill; here it is the combers (2.3 → 3 machines gives 31% headroom, but a noil increase to 18% would raise lap duty and consume it).
- **Draft and count chain check:** the counts at each stage (card sliver 0.15 Ne → lap 60 g/m → comber sliver 3.9 g/m → drawframe 3.84 g/m → roving 1.20 Ne → yarn Ne 30) must multiply out through the drafts of Topics 2–5 without gaps.
- **Revision:** a spin plan is recalculated whenever the count, blend, noil level or target changes — it is a living document, and process control (YM-4101) exists to keep the real mill on the plan's numbers.

Glossary

Spin plan — the numerical design linking product, material flow and machine numbers for a spinning mill. **Retention** — the fraction of feed leaving a stage as its main product (100 − waste%). **Yarn realisation** — yarn produced as a percentage of raw material issued. **Duty** — the daily mass a department must process in the plan. **Bottleneck** — the stage with least spare capacity, which limits mill output. **Installed capacity** — what the rounded-up machine numbers can actually produce per day.

Key points

• Plan backwards: yarn target ÷ stage retentions, stage by stage, gives every stage's feed.

• For combed Ne 30 at 2000 kg/day, mixing = 2732 kg/day and realisation = 73.2% — the comber's 15% noil dominates.

• Machine numbers come from the production rates of Topics 1–5: 1 blowroom line, 8 cards, 3 combers, 2 drawframe deliveries per passage, 2 simplex frames, 5 ring frames (6000 spindles), 80 winding drums.

• Always round up and check the balance stage by stage; spare capacity is a design feature, not waste.

• A change in count, noil or target reopens the whole calculation.

Practice

1. A carded Ne 20 mill targets 1500 kg/day with an overall realisation of 88%. Calculate the daily mixing requirement.
2. Using the ring data of the answer-key example in Topic 5 of this course (Ne 24 needs 24.9 g/h per spindle), calculate the spindles needed for 1200 kg/day of Ne 24 at 24 h and 92% efficiency, and the number of 1200-spindle frames.
3. In the plan above, noil is raised from 15% to 18% (comber waste becomes 18.5%, retention 0.815). Recalculate the comber lap duty and the mixing requirement, keeping all other retentions unchanged.

Answer key

1. Mixing = 1500 / 0.88 = **1704.5 kg/day**.
2. Per spindle per day = 24.9 × 24 × 0.92 = 549.8 g. Spindles = 1,200,000/549.8 = **2183 spindles** → 2183/1200 = 1.8 → **2 frames (2400 spindles)**.
3. Comber feed = 2050.8 / 0.815 = **2516.3 kg/day** of lap (was 2427.0). Working back: card sliver = 2516.3/0.995 = 2529.0; blowroom lap = 2529.0/0.955 = 2648.2; mixing = 2648.2/0.935 = **2832.3 kg/day** (was 2731.7 — about 100 kg/day more cotton for the same yarn, the cash cost of the extra noil).